V5 vs V6 Slot Assignment
Evidence-Backed Technical Verdict

Revised Analysis | Rate-Limiter Service | Principal Distributed-Systems Review

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Overall Winner: V5 for production workloads at 500 TPS / 24 pods.

V5's synchronous counter UPDATE is lightweight enough at 500 TPS that the contention concern does not materialize as a bottleneck (2.6% row-lock utilization, 0.02ms average wait). V6 introduces a background scheduler with an expensive COUNT(*) GROUP BY MERGE query that runs ~8 times/second across the cluster, adding DB load and operational complexity without clear throughput benefit at this scale.

When V6 becomes preferable:

- > 2,000 TPS sustained with extreme concentration on a few windows (utilization approaches 50%+ on counter row locks)
- Counter row becomes a measured bottleneck in production AWR/ASH reports (not theoretical)
- Scheduler is acceptable as an operational dependency and Quartz migration is not planned

B) Scorecard

#	Parameter	V5	V6	Winner	Why	Conf.
1	Global rate limiting	Shared counter, immediate	Shared counter, eventually consistent	Tie	Both use global WNDW_STRT_TS-keyed counters	High
2	Proximity to requestedTime	Fresh counters, accurate picker	Stale counters, mis-picks, soft guard re-pick	V5	V5 picker has real-time occupancy	Med
3	Correctness at target workloads	Gap design absorbs concurrency	Soft guard compensates for stale counters	V5	V5 doesn't need secondary mechanism	High
4	Performance at high TPS	3 DB calls; lock ~0.02ms avg wait	4 DB calls; no lock but +1 COUNT	V5	Counter lock negligible at 500 TPS (2.6% util.)	High
5	DB load at high TPS	N INSERTs + N simple UPDATEs	N INSERTs + N COUNTs + ~8 heavy MERGEs/sec	V5	V6 replaces cheap UPDATEs with COUNTs + heavy MERGEs	Med
6	Maintainability	Self-contained; no background process	Scheduler + MERGE SQL + 2 extra indexes	V5	V6 adds significant operational surface	High
7	Multi-pod (24 pods)	Serialize briefly on counters (~0.02ms)	Insert independently; 24 schedulers run MERGEs	V5	Counter serialization trivial; 8 MERGEs/sec is not	Med
8	Quartz scheduler impact	N/A	Staleness 25x worse, picker accuracy drops	V5	V5 has no scheduler dependency	High
9	Failure modes	Crash-consistent (single txn)	Scheduler failure = stale counters	V5	V5 has no background dependency	High
10	Consistency model	Immediate	Eventually consistent (~125ms)	V5	Counter always accurate in V5	High
11	Observability	Counter table = ground truth	Counter table = lagging indicator	V5	Operators query RL_WNDW_CT directly	Med
12	Retry / rollback	Not needed (gap absorbs concurrency)	Soft guard loop re-picks on misdirection	V5	Capacity model designed so retry unnecessary	High
13	Hotspot contention	Counter row: 2.6% utilization	No hot-path row locks	V6	V6 eliminates counter locks (trivial at 500 TPS)	Med
14	Idempotency under race	UNIQUE(EVENT_ID) + conditional skip	UNIQUE(EVENT_ID) + no counter write	Tie	Both correct	High
15	Scheduler resilience	N/A	Soft guard catches stale picks	N/A	Only applies to V6	Low
16	Throughput ceiling	Lock saturation ~1,250 TPS/window	Bounded by INSERT throughput	V6	At >2000 TPS, V6 ceiling higher	Med

Tally: V5 wins 10, V6 wins 2, Tie 2, N/A 2

C) Deep Dive Sections

C1. Global Rate Limiting

Both versions share an identical global model: `RL_WNDW_CT` is keyed solely by `WNDW_STRT_TS`, not by caller, requestedTime, or configName.

V5 (`SlotAssignmentServiceV5.kt:218-219`): `upsertCounter(windowStart)` atomically increments the counter within the INSERT transaction.

V6 (`SlotAssignmentServiceV6.kt:238-258`): INSERT-only in hot path. `WindowCounterRefreshScheduler.kt:39-40` reconciles via `MERGE INTO RL_WNDW_CT` (`WindowSlotCounterRepository.kt:232-252`).

Both enforce a single global rate limit per window. **Tie** -- the difference is timing (immediate vs eventual), covered in C10.

C2. Proximity of scheduledTime to requestedTime

Both use identical `WindowPicker.pickProximityWeightedRandom()` (`WindowPicker.kt:25-46`):

```
weight = max(0, threshold - occupancy) x (rangeSize - index)
```

V5 (`SlotAssignmentServiceV5.kt:117`): `readOccupancy()` returns the real-time counter (updated inline per INSERT at line 218-219). The picker sees current fill levels and preferentially routes to the closest window with genuine remaining capacity.

V6 (`SlotAssignmentServiceV6.kt:127`): `readOccupancy()` reads stale counters (updated only by the background scheduler every ~125ms effective). At 500 TPS, ~62.5 events arrive between refreshes, spread across ~30 windows = ~2.1 stale events/window. The picker may assign a window that appears to have capacity but is actually closer to full. The soft guard (`SlotAssignmentServiceV6.kt:213-216`) then rejects this pick and the re-pick loop selects a **farther** window.

V5 wins on proximity -- the soft guard is a correction mechanism, not an improvement.

C3. Correctness at Target Workloads

V5's Capacity Model: The Gap IS the Concurrency Buffer

V5's design is built around the configurable softMax/maxSlotsPerWindow gap: `softMax = floor(900 x 90 / 100) = 810` (`SlotAssignmentServiceV5.kt:69`), `maxSlotsPerWindow = 900` (`application.yaml:139`). **Gap = 90 slots = 10% headroom.**

Worst-case overshoot calculation: 24 pods, each sending ~21 TPS. With proximity weighting and 30 windows per chunk, the hottest window receives ~6.5% of traffic = ~32.5 TPS (`WindowPicker.kt:31-34`). If 24 threads all read occupancy at the same instant when the window shows count=809, all 24 insert. Result: 809 + 24 = 833 slots. **Still below maxSlotsPerWindow (900).** The gap absorbs this entirely.

Phase 2 uses `maxSlotsPerWindow` (900): worst-case 899 + 24 = 923 slots. At 30s: 923/30 = 30.77 TPS. Target range: 27-33 TPS. **Within tolerance.** V5 does not need a soft guard, retry, or rollback. The user can tune `soft-max-percent` to accommodate any pod count.

V6's Correctness Dependency on Soft Guard

V6 has the same racing problem, but because counters are stale, the racing window is wider. The soft guard (`SlotAssignmentServiceV6.kt:213-216`) catches gross errors but has its own race: between `countSlotsInWindow()` (line 213) and `insertEventSlot()` (line 240), other threads can insert. Worst case: 899 + 24 = 923, same as V5.

V5 wins on correctness: its capacity model is self-contained. V6's soft guard exists to compensate for a problem V6 created (stale counters), and the final overshoot bound is the same.

C4. Performance at High TPS -- Counter Lock Math

V5 counter UPDATE contention -- M/M/1 queueing analysis

Transaction structure (`SlotAssignmentServiceV5.kt:207-231`): (1) `insertEventSlot()` ~0.5ms, (2) `upsertCounter()` ~0.3ms, (3) COMMIT ~0.5ms. Counter row lock hold time: ~0.8ms.

Steady-state scenario (30 windows available):

Metric	Value
Hottest window share (W0)	30/465 = 6.5%
Arrival rate (lambda)	500 x 0.065 = 32.5 req/s
Service rate (mu)	1 / 0.0008 = 1,250 req/s
Utilization (rho)	32.5 / 1,250 = 2.6%
Average wait in queue	0.021ms
P99 wait	0.78ms

Transient worst-case (3 windows remain):

Metric	Value
Hottest window share	50%
Arrival rate (lambda)	500 x 0.5 = 250 req/s
Utilization (rho)	250 / 1,250 = 20%
Average wait in queue	0.20ms
P99 wait	~3.0ms
Duration of this state	~1.2 seconds (until windows fill)

Hot-path latency comparison:

Version	Breakdown	Total
V5	Skip ptr 0.1ms + Occupancy 0.5ms + INSERT+UPDATE+COMMIT 1.3ms + 0.02ms lock	~1.92ms
V6	Skip ptr 0.1ms + Occupancy 0.5ms + Soft guard COUNT 0.3ms + INSERT 0.5ms	~1.4ms

> **Conclusion**: At 500 TPS, V5's counter contention is **not a bottleneck**. V5's 3 DB calls with trivial lock wait outperform V6's 4 DB calls in total transaction cost. The extra round-trip for the soft guard COUNT costs more than the counter lock wait. **V5's counter lock becomes problematic above ~1,250-2,000 TPS sustained.**

C5. DB Load at High TPS

V5 at 500 TPS for 1 hour (1.8M events):

Operation	Count	Type	Cost/op
Skip pointer READ	1.8M	PK lookup	~0.1ms
Occupancy READ	1.8M	Range scan (30 rows)	~0.5ms
Slot INSERT	1.8M	Single-row INSERT	~0.5ms

Operation	Count	Type	Cost/op
Counter UPDATE	1.8M	Single-row PK UPDATE	~0.3ms
Counter INSERT (cold)	~2,000	Single-row INSERT	~0.5ms
Skip pointer INSERT	~120	Single-row INSERT	~0.3ms

Total: ~5.4M DB ops (3/request). All writes are simple single-row PK operations.

V6 at 500 TPS for 1 hour (1.8M events):

Operation	Count	Type	Cost/op
Skip pointer READ	1.8M	PK lookup	~0.1ms
Occupancy READ	1.8M	Range scan (30 rows)	~0.5ms
Soft guard COUNT(*)	1.8M+	Index scan per window	~0.3ms
Slot INSERT	1.8M	Single-row INSERT	~0.5ms
Skip pointer INSERT	~120	Single-row INSERT	~0.3ms
Scheduler MERGE	~28,800	Subselect + COUNT GROUP BY + upsert	~10-30ms

Total: ~7.2M hot-path ops (4/request) + 28,800 heavy MERGE statements.

Scheduler MERGE breakdown per execution:

Metric	Value at 500 TPS / 24 Pods
MERGE executions	~8/second
Rows scanned per MERGE (inner: CREAT_TS index)	~3,000 (6s x 500 TPS)
Rows scanned per MERGE (outer: COUNT GROUP BY)	~45,000 (50 windows x 900 rows)
Total logical reads from slot table	~360,000/second
Connection hold time per MERGE	~10-30ms
Connections occupied by schedulers	~1 avg, ~2-3 during overlap
Row locks on counter table	50 rows x 8/sec x 20ms = 8,000 row-lock-ms/sec

V5 is lighter on DB load. V5's 1.8M simple UPDATEs total ~540s cumulative execution time. V6 replaces these with 1.8M COUNTs (~540s) PLUS 28,800 heavy MERGEs (~576s at 20ms each). V6's total DB work is higher.

C6. Maintainability / Operational Complexity

V5: Self-contained. No background process. No scheduler to monitor. No MERGE SQL to debug. No index dependencies beyond standard PK indexes. Config surface: 7 parameters.

V6 adds:

- `WindowCounterRefreshScheduler` (`WindowCounterRefreshScheduler.kt:26-45`): must be monitored
- MERGE SQL (`WindowSlotCounterRepository.kt:232-252`): 15-line raw SQL with nested subquery
- Two extra indexes: `RL_EVENT_SLOT_DTL_I01X(WNDW_STRT_TS)` and `I02X(CREAT_TS)`
- Two extra config parameters: `counter-refresh-every`, `counter-refresh-since`
- New failure modes: scheduler crash, scheduler lag, MERGE contention, index bloat
- Monitoring requirements: scheduler health, counter staleness, MERGE execution time

C7. Multi-Pod Behavior (24 Pods)

V5: 24 pods each do INSERT + UPDATE. The UPDATE serializes on counter rows with ~2.6% utilization (steady state) and ~20% utilization (transient). P99 addition: 0.78ms steady / 3ms transient. Manageable.

V6: 24 pods insert independently (zero write contention in hot path). However, 24 pods each run `WindowCounterRefreshScheduler` every 3s = ~8 MERGE/sec, each scanning ~45K rows. The scheduler itself creates contention -- not in the hot path, but in background load.

V5 distributes light work across all requests; V6 concentrates heavy work into periodic bursts.

C9. Additional Parameters

C9.1 Failure Modes

V5: INSERT + UPDATE in same `transaction {}` (`SlotAssignmentServiceV5.kt:207`). Both commit or both rollback. Crash-consistent by construction. Counter drift impossible. No background dependencies.

V6: Scheduler failure (`WindowCounterRefreshScheduler.kt:41-42`) = counters freeze. At 10s outage: ~5,000 unaccounted slots, ~29 stale/window, picker misdirects ~3.6%. CREAT_TS index missing = full table scan, cascading staleness. Scheduler-DB connection leak at 8 MERGEs/sec with 5s timeout = 40 connections consumed during DB slowdown.

C9.2 Consistency Model

V5: Strongly consistent. Every committed slot has its counter incremented atomically in the same transaction.

V6: Eventually consistent (~125ms with 24 pods). Picker makes decisions on stale data, relies on soft guard as correction layer.

C9.3 Observability

V5: `SELECT WNDW_STRT_TS, SLOT_CT FROM RL_WNDW_CT` returns ground truth. **V6**: Counter table lags by up to 6s; operators must query slot table with expensive GROUP BY for real-time data.

C9.4 Retry / Rollback Behavior

V5: No retry needed. Phase 1 overshoot: 810 + 24 = 834 (below 900). Phase 2 overshoot: 900 + 24 = 924 = 30.8 TPS (within 27-33 TPS tolerance). Gap is tunable via `soft-max-percent`.

V6: `tryClaimWithSoftGuard()` loop is a retry mechanism for stale-counter mis-picks. Each retry costs 1 COUNT query (~0.3ms). Exists because V6's stale counters create the problem.

C9.5 Hotspot Contention

V5: Steady state 2.6% utilization, 0.02ms avg wait. Transient 20%, 3ms P99. Non-issue at 500 TPS. **V6**: Zero hot-path locks but scheduler MERGEs create 16% background utilization on counter rows.

C9.6 Scheduler DB Impact (V6 Only)

360,000 logical reads/second from slot table. ~1 connection always occupied. Buffer cache pressure from wide-range MERGE scans competing with hot-path INSERT/COUNT. V5 has zero background DB work.

D) Practical Worst-Case Scenarios

V5 Worst Cases

D1. Transient Counter Lock Spike During Chunk Exhaustion

Trigger: Chunk nearly exhausted, 3 windows remain. Proximity weighting concentrates 50% of traffic. **Math**: lambda=250 req/s, rho=20%, P99 wait ~3ms. **Duration**: ~1.2 seconds until windows fill and skip pointer advances. **Impact**: P99 spike of ~3ms for ~1.2s, once per chunk (~every 15 minutes). Fully absorbed.

D2. Phase 2 Full-Range Scan

Trigger: All Phase 1 chunks exhausted at softMax. Phase 2 reads 960 windows at once. **Impact**: One 2-3ms read. Requires 777,600 events before triggering. Rare under 1-hour load. Cite: `SlotAssignmentServiceV5.kt:137-148`.

D3. Skip Pointer Stale Read Under Multi-Pod Race

Trigger: Two pods read skip pointer simultaneously, both scan same chunk. **Impact**: One wasted occupancy read (~0.5ms). Duplicate INSERT caught by composite PK. Cite: `SkipPointerRepository.kt:51-64`.

V6 Worst Cases

D4. Scheduler Outage -- Cascading Staleness

Trigger: Scheduler crashes on all pods. **Timeline**: T+3s: 1,500 stale events, 5% misdirection. T+10s: 5,000 stale events, ~2 re-picks/request. T+30s: 15,000 stale events, every request triggers Phase 2 (full-range rescan with soft guard COUNT per pick = heavy DB load). **Impact**: Latency degrades from ~1.4ms to ~5-10ms. DB load increases ~3x. Cite: `WindowCounterRefreshScheduler.kt:41-42`.

D5. 24-Pod Soft Guard Race -- Window Overshoot

Trigger: 24 pods all call `countSlotsInWindow(W)` within 1ms, all see count=899. All 24 pass guard, all INSERT. Window gets 923 slots = 30.77 TPS. **Note**: Same overshoot bound as V5. Soft guard does not eliminate it. Cite: `SlotAssignmentServiceV6.kt:213` and `:240`.

D6. MERGE Contention During Scheduler Overlap

Trigger: Two scheduler instances execute MERGE simultaneously. Oracle serializes them. **Impact**: Second MERGE waits ~20ms. At 8/sec: 16% overlap probability. If MERGE takes >3s, `ConcurrentExecution.SKIP` prevents per-pod overlap but not cross-pod. Cite: `WindowSlotCounterRepository.kt:232-252`.

E) Quartz vs Quarkus Scheduler Impact

Current State: Quarkus @Scheduled (Per-JVM)

Metric	Value
Effective refresh interval	3s / 24 = ~125ms
Counter staleness (events)	500 x 0.125 = ~62.5 total
Per-window staleness	62.5 / 30 = ~2.1 events
Picker mis-pick rate	2.1 / 810 = ~0.26%
Scheduler MERGE executions	~8/sec
Scheduler DB load	~360K logical reads/sec

Hypothetical: Quartz Scheduler (Cluster-Singleton)

Metric	Quarkus (24 pods)	Quartz (1 instance)	Delta
Effective refresh interval	~125ms	3,000ms	24x worse
Counter staleness per window	~2 events	~50 events	25x worse
Picker mis-pick rate	~0.26%	~6.2%	24x worse
Extra DB calls from re-picks	~0.005/req	~0.12/req	Noticeable
P99 latency impact	~0ms	~1-2ms	Measurable
MERGE executions	~8/sec	~0.33/sec	24x less DB load
Overshoot bound per window	~24 (threads)	~24 (threads)	Same

Key insight: Overshoot bound is determined by concurrent threads, not counter staleness. Quartz degrades picker accuracy (more mis-picks) but does not change the overshoot ceiling.

V5 under Quartz: Zero impact. No scheduler dependency.

F) Final Recommendation

Overall Best Default: V5 for the stated workloads (500 TPS, 24 pods, production Oracle). V5 is simpler, has fewer moving parts, provides immediate consistency, and its counter contention is negligible at 500 TPS (2.6% utilization, 0.02ms average wait). V6's added complexity (scheduler, MERGE, extra indexes, soft guard) solves a problem that doesn't exist at this scale.

Best per Parameter

Parameter	Best	Notes
Global rate limiting	Tie	Identical model
Proximity to requestedTime	V5	Fresher counters = better initial picks
Correctness	V5	Gap model absorbs concurrency by design
Performance at high TPS	V5	3 calls + 0.02ms lock < 4 calls
DB load at high TPS	V5	Simple UPDATEs < COUNTs + heavy MERGEs
Maintainability	V5	Self-contained, no scheduler
Multi-pod (24 pods)	V5	Lightweight lock < background MERGE load
Quartz scheduler	V5	No dependency
Failure modes	V5	Crash-consistent, no background processes
Consistency model	V5	Immediate
Observability	V5	Counter table = truth
Retry behavior	V5	Doesn't need it
Hotspot contention	V6	Zero hot-path locks (trivial at 500 TPS)
Idempotency	Tie	Both correct
Throughput ceiling	V6	Higher theoretical ceiling (>2000 TPS)

Decision Matrix by Priority

Priority	Rec.	Rationale
Strict correctness-first	V5	Immediate-consistency counters; gap model; no secondary mechanism
Throughput-first (500 TPS)	V5	3 DB calls < 4; counter lock negligible; no MERGE overhead
Throughput-first (>2000 TPS)	V6	Counter lock utilization exceeds 50%; INSERT-only scales better
Ops-simplicity-first	V5	No scheduler, no MERGE, no extra indexes
Quartz-migration planned	V5	V6's advantage requires per-JVM scheduler

G) Gaps in Current Tests and Benchmarks Needed

Test Gaps

- **No latency benchmark**: Neither suite measures P50/P99 under load. Tests verify "no deadlocks" but don't capture timing distributions.
- **No counter-lock contention measurement (V5)**: `SlotAssignmentServiceV5Test.kt:421-468` validates correctness but doesn't measure UPDATE wait time at 20+ threads.
- **No scheduler-failure test (V6)**: No test stops the scheduler and verifies behavior under stale counters. Should verify no window exceeds maxSlotsPerWindow + bound.
- **V6 over-allocation bound too loose**: `SlotAssignmentServiceV6Test.kt:568-575` asserts count <= maxSlots + threadCount (4+20=24). A tighter bound like maxSlots + 5 would better validate soft guard.
- **No multi-pod simulation**: All tests run single-JVM. Skip pointer coordination untested for multi-writer with network latency.
- **No long-horizon sustained test**: Neither tests 500K events at +30 days. Need 1000 events at 100 TPS targeting +30 days.
- **No MERGE performance test (V6)**: Scheduler tests use 8 slots. Need 3000 slots across 50 windows.
- **No Phase 2 concurrent performance test**: Phase 2 reads 960 windows. Need 50 concurrent Phase 2 requests.

Benchmarks Needed

- **V5 counter-lock P99**: JMH or Gatling, 500 TPS, 24 threads, real Oracle 19c. Validate M/M/1 model.
- **V6 MERGE execution time at scale**: 50 windows x 900 slots under 500 TPS INSERT load.
- **V6 soft guard rejection rate vs staleness**: Vary counter-refresh-every 1s-10s, plot rejection rate.
- **End-to-end comparison**: 500 TPS, 10 min, 24 threads, real Oracle. P50/P95/P99 + DB utilization.
- **Quartz-mode V6**: Single scheduler thread, 500 TPS. Measure staleness and rejection rate.
- **Buffer cache impact (V6)**: Measure buffer cache hit ratio with/without scheduler MERGEs.